

Assignment # 1

Government Jinnah Islamiye graduate
College Prialkot

Title:-

Applied Chemistry (321)

Topic

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Roll no

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Short Questions

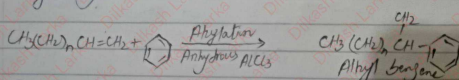
(i)

Linear Alkylbenzenes:-

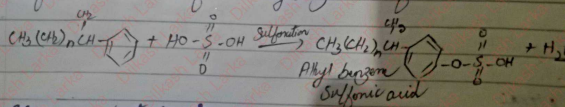
Ans:- The formation of linear alkylbenzene (LAB) and its sulfonation using to produce linear alkylbenzene sulfonate (LAS), which is a detergent.

Step 1: Formation of an organic acid

A long-chain alkene reacts with benzene to form alkylbenzene.

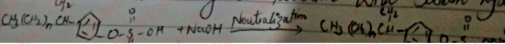


The alkylbenzene formed is then reacted with concentrated sulfuric acid to form alkylbenzene sulfonic acid.



Step 2: Neutralization

The alkylbenzene sulfonic acid is then converted to a sodium salt by a reaction with sodium hydroxide.



(ii)

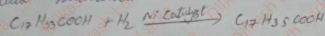
Two applications of hydrogenation reactions-

Pro:- Manufacturing of margarine and vegetable ghee:

Hydrogenation of vegetable oils (unsaturated fats) converts them into solid or semi-solid fats like margarine and vanaspathi ghee.

Example reactions:-

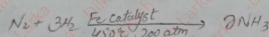
Unsaturated fats + H_2 $\xrightarrow{Ni\ catalyst}$ Saturated fat
For instance, oleic acid (unsaturated) is hydrogenated to stearic acid (saturated).



Production of Ammonia (Haber process):

Hydrogenation of nitrogen gas (N_2) with hydrogen (H_2) under high temperature and pressure produces ammonia (NH_3), which is used in fertilizers.

Example reactions:-



(iii)

Cullet:- Cullet refers to waste or broken glass used as a raw material in glass production. It is melted and reused to manufacture new glass products, reducing energy consumption and the need for raw materials like silica sand.

Uses:-

1. Glass manufacturing: Cullet is melted and reused to make new glass products. It helps in making bottles, jars, windows, and fiberglass. Using cullet reduces the need for raw materials like sand. This makes the glass industry more energy efficient.

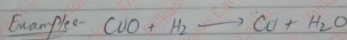
2. Energy Conservation: Cullet melts at a lower temperature than raw materials. This reduces fuel consumption in glass production.

3. Waste Reduction: Recycling cullet prevents glass from ending up in landfills. It reduces environmental pollution and waste problems.

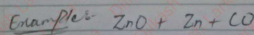
(iv)

Two Reducing agent:-

1. Hydrogen (H_2): Hydrogen is a strong reducing agent commonly used in industrial processes, such as the hydrogenation of alkenes and the reduction of metal oxides to pure metals.



2. Carbon (C): Carbon, in the form of coke or charcoal, is widely used as a reducing agent in metallurgy to extract metals from their ores.



Definition:-

A reducing agent (or reductant) is a substance that donates electrons in a redox reaction, causing another substance to be reduced while itself gets oxidized.

(vi)

Two examples of unit operations:-

Definition: A unit operation is a basic step in an industrial process that involves a physical change, such as separation, mixing, or heat transfer, without altering the chemical composition of substances. These operations are essential in chemical, food and pharmaceutical industries to process raw materials into usable products.

Examples of unit operations: (i) Evaporation (ii) Filtration

(i) **Evaporation:** A process where a liquid is heated to remove a volatile solvent (usually water), leaving a concentrated solution. It is widely used in industries like food processing (e.g., concentrating milk) and chemical manufacturing (e.g., salt production).

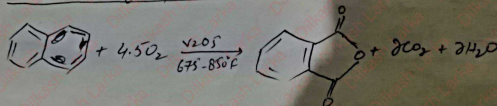
(ii) **Filtration:** This unit operation separates solid from a liquid or gas by passing the mixture through a porous medium that retains the solids while allowing the fluid to pass through. It is commonly used in water treatment plants to remove suspended particles from water, ensuring it is clean for drinking or industrial use, by passing the water through sand filters or membrane systems.

(vii)

Structure of Phthalic anhydride:

Phthalic anhydride is the organic compound with the formula $C_8H_4O_3$. It is the anhydride of phthalic acid. On large scales, phthalic anhydride is produced by oxidation of naphthalene and ortho-xylene.

From oxidation of naphthalene:



Applications:

1. Phthalic anhydride is used to manufacture phthalate esters (e.g., Diethyl phthalate, DEP), which are added to PVC plastics to improve flexibility and durability.
2. It is a key raw material in the production of unsaturated polyester resins, which are used in fiberglass, coatings, and composite materials (e.g., car body parts and boats).
3. It is used in the production of phthalocyanine dyes and pigments, which are widely applied in inks, paints, and textiles.
4. It is used as an intermediate in the synthesis of certain drugs, such as thalidomide and anti-inflammatory medicines.

(viii)

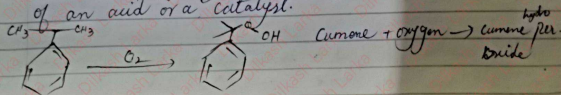
Cumene hydroperoxide:

Cumene hydroperoxide ($C_9H_{10}O_2$) is an organic peroxide formed by the oxidation of cumene (isopropylbenzene). It is an important intermediate in the industrial production of phenol and acetone.

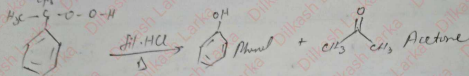
Preparation of cumene:

Step # 1

Cumene hydroperoxide is prepared by the oxidation of cumene (isopropylbenzene) with oxygen (O_2) in the presence of an acid or a catalyst.



This hydroperoxide is then treated with diluted HCl (or H₂SO₄) at a controlled temperature, leading to cleavage, which produces Phenol (C₆H₅OH) and acetone (CH₃COCH₃).



Applications:-

1. It decomposes into Phenol (C₆H₅OH) and acetone (CH₃COCH₃) using acidic catalysts.
2. It is used in the plastic and rubber industry to start polymerization reactions.
3. A key reagent in the synthesis of various organic peroxides and other chemicals.

(viii)

Butadiene at atmospheric conditions:-

At atmospheric conditions (room temperature and pressure), butadiene (C₄H₆) exist as a colorless gas.

Explanation:-

Butadiene (C₄H₆) is a gas at room temperature and atmospheric pressure. It has a boiling point of -4.4°C (24.8°F), meaning it can easily liquify under slight cooling or increased pressure. It has a mild, somewhat sweet, or gasoline-like odor and is highly flammable and reactive. Its molecular formula is (C₄H₆), and its structure consists of a conjugated diene with the formula CH₂=CH-CH=CH₂.

Uses:-

1. A key monomer in making synthetic rubbers like Styrene-butadiene rubber (SBR) and polybutadiene rubber (PBR). These rubbers are used

in tires, footwear, and industrial belts.

2. It is used in producing acrylonitrile-butadiene-styrene (ABS) plastic, known for its strength and toughness. Also used in latex and resins.
3. Acts as a raw material for producing adiponitrile (for nylon) and chloroprene (for neoprene rubber).

(ix)

Glass Chemically:-

Chemically glass is defined as an amorphous (non-crystalline) solid material made by cooling a molten mixture of inorganic compounds, primarily silica (SiO₂), along with other oxides like sodium oxide (Na₂O) and calcium oxide (CaO). The rapid cooling prevents the formation of regular crystalline structures, resulting in a random atomic arrangement that gives glass its unique properties.

1. Primary components: The main component of glass is silica (SiO₂), which forms a continuous network of silicon and oxygen atoms.
2. Network structure: Each silicon atom (Si) is tetrahedrally bonded to four oxygen atoms (O). The SiO₄ tetrahedra are linked by sharing oxygen atoms, forming a three-dimensional network.
3. Modifiers: Sodium oxide:- Sodium ions (Na⁺) are added to the silica network to lower the melting point and improve workability.

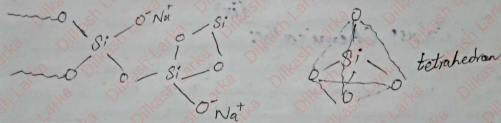
Calcium oxide (CaO): Enhances chemical durability and mechanical strength.

iv. Amorphous Nature: Unlike crystalline solids, glass lacks a long-range ordered structure. The arrangement of atoms is disordered, which is why glass is considered an amorphous solid.

General chemical representation:

SiO_2 (70-75%) + Na_2O (12-16%) + CaO (8-12%) + Minor Additives
for soda lime glass (the most common type).

Structure:



(x) General formula of Glass:

The general formula of glass, particularly for soda-lime glass (the most common type) is:
 $\text{SiO}_2 \cdot \text{Na}_2\text{O} \cdot \text{CaO}$

- (i) SiO_2 (Silicon dioxide, 70-75%, It forms the glass network.
- (ii) Na_2O (Sodium oxide, 12-16%, Acts as a flux, reducing the melting point.
- (iii) CaO (Calcium oxide, 5-10%, It increases durability and chemical resistance.

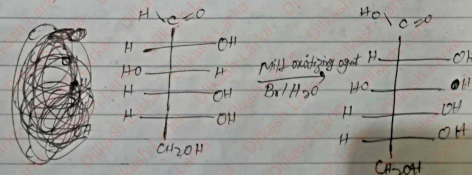
Additional components may include:

- MgO (Magnesium oxide, 1-5%, enhances durability
- Al_2O_3 (Aluminium oxide, 1-3%, Improves chemical resistance
- K_2O (Potassium oxide 1-2%, Affects thermal expansion.

The exact composition varies depending on the intended use (e.g., window glass, bottles, laboratory glassware).

(xi) Which one acid is produced by oxidation of Carbohydrates:

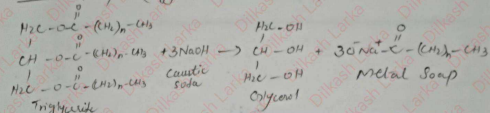
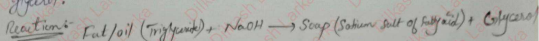
Amino acid (e.g., Gluconic acid) is produced by the oxidation of Carbohydrates, specifically glucose. In this process, the aldehyde ($-\text{CHO}$) group at the first carbon of glucose is oxidized to a carboxyl ($-\text{COOH}$) group. This reaction is commonly catalyzed by the enzyme glucose oxidase or through chemical oxidation. Gluconic acid is widely used in the food and pharmaceutical industries.



(xii) Saponification Process:

Saponification is the chemical reaction in which fats or oils (triglycerides)

react with a strong base (NaOH or KOH) to produce soap and glycerol.



uses: (i) It is the main process used to manufacture soaps for cleaning and hygiene.

(ii) Some detergents are made using a similar process for washing clothes and dishes.

(iii) It is used in converting fats and oils into biodiesel fuel.

(iv) It helps in making creams, lotions, and other skincare products.

(xii)

Speciality of Pyrex glass:

Pyrex glass is made up to 80-82% silica (SiO_2), 20% Boron trioxide (B_2O_3), 10% Sodium oxide (Na_2O), and alumina (Al_2O_3). It has a low coefficient of thermal expansion (CTE), which makes it resistant to temperature changes and less prone to thermal shock. Pyrex can withstand high temperatures and sudden cooling without cracking. It is highly resistant to acids, bases, and other chemicals, making it ideal for laboratory glassware. Pyrex is commonly used for chemical glassware, cooking glass, car headlamps, reagent bottles, optical components, and household cookware.

(xiv)

Raw materials for colored glass:-

Certain metallic oxides or salts, when added to molten glass,

produce different shades of colors. These compounds are used in the manufacturing of colored glass. The key raw materials used for coloring glass are:

1. Red Glass: Cadmium oxide (CdO) (2-4%), Selenium (Se) (2%).
2. Ruby Red Glass: Colloidal Gold, copper(II) oxide (Cu_2O).
3. Light yellow glass: Cerium oxide (CeO_2) (2%), Titanium oxide (7%).
4. Deep Blue Glass: Cobalt oxide (CoO) (0.2%).
5. Greenish Blue glass: Copper(II) oxide (CuO).
6. Emerald Green glass: Chromium oxide (Cr_2O_3) (0.25%), Cobalt oxide (CoO) (0.002%).
7. Amber Glass: Iron Sulfide (FeS) (0.3%).
8. Black Glass: Manganese Dioxide (MnO_2) + Ferric oxide (Fe_2O_3).
9. Yellow glass: Cadmium Sulfide (CdS).

These materials influence the final color of the glass by interacting with the molten mass during production.

(xv)

zwitter ionic surfactants:

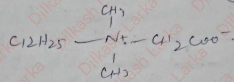
Zwitterionic and amphoteric surfactants are surfactant molecules that exhibit both anionic and cationic dissociation, it is called zwitter or amphoteric.

Examples:

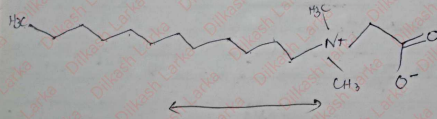
Synthetic compounds like betaines and sulfobetaines, as well as natural substances like amino acids and phospholipids. Some of these surfactants change

their charge depending on pH - being cationic at low pH, anionic at high pH, and amphoteric at intermediate pH. Due to ^{this} high biological compatibility and low toxicity, they are mainly used in specialized applications like cosmetics. Commonly used surfactants:-

1. Dodecyl Betaine $\rightarrow$ It has both positive ($-N^+(H_3)$) and negative ($-COO^-$) charges in the same molecule, making it a zwitter ionic surfactant.



2. lauryl Betaine $\rightarrow$ lauryl Betaine is a zwitterionic surfactant with both hydrophilic ($-COO^-$, $-N^+(CH_3)_2$) and hydrophobic ($C_{12}H_{25}$) parts.



①

Long Questions

(i)

Multiple effect Evaporators

Multiple effect evaporators are a series of evaporators connected in sequence, where the vapor produced in one effect is reused as the heating medium for the next. Each successive effect operates at a lower pressure and temperature, allowing efficient heat utilization and reduced steam consumption. This method is widely used in industries for concentrating solutions and minimizing energy costs.

Why Multiple Effect Evaporators are preferred Single

(i) Energy Efficiency:-

* In a single-effect evaporator, steam is used once and discarded. In a multiple-effect system, the vapor from one effect is reused to heat the next.

Example:- In a triple-effect evaporator, steam heats the first effect, the vapor from the first effect heats the second, and so on. Each effect operates at a lower pressure and temperature, ensuring better energy utilization.

* This reduces steam consumption by 50-80%, leading to significant energy savings.

(ii) Lower operating costs:-

By requiring less fresh steam, these systems significantly reduce fuel and operational costs.

(iii) Higher Capacity:-

Handles larger volumes of liquid due to its multi-effect design.

design, making it ideal for industrial scales. (2)

(iii) Better suitability of Heat Sensitive Materials

The lower temperatures in later effects prevent thermal degradation of sensitive substances e.g., dairy products, pharmaceuticals.

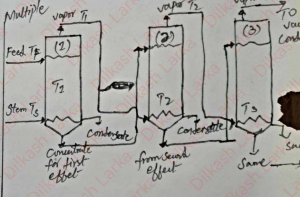
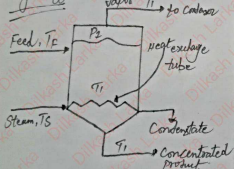
(iv) Environmental Benefits

Reduced steam usage means lower fuel consumption and fewer greenhouse gas emissions.

(v) Scalability and Flexibility

Multiple-effect systems can be customized with two, three, or more effects depending on process requirements, offering greater flexibility for industries.

Single effect:-



Applications (64) page number

(ii)

Describe role of Borax and Soda in manufacturing glass.

In glass manufacturing, Borax (Sodium Borate $\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$) and Soda Ash (Sodium carbonate, Na_2CO_3) act as essential fluxing agents. They lower the melting point of silica (SiO_2), making the glass formation process more efficient. Additionally, they improve the strength, durability, and thermal resistance of the final glass product.

Role of Borax

(i) Reduction in Melting point:- Borax lowers the melting temperature of Silica

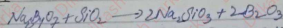
(3)

making it easier to form glass at lower energy costs.

(ii) Improvement in thermal and chemical resistance:- It enhances the ability of glass to withstand temperature changes and resist chemical corrosion, making it useful for laboratory glassware and heat-resistant cookware (e.g., Pyrex).

(iii) Increase in Durability and Strength:- The addition of borax makes the glass more durable and less likely to break under stress.

Reaction:-



Borax reacts with silica to form sodium silicate and boron oxide. These compounds improve the stability, durability and thermal resistance of glass.

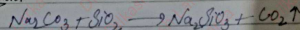
2. Role of Soda ash

(i) Lowering of melting point:- Soda ash reduces the melting point of silica from around 1700°C to $1500-1550^\circ\text{C}$, making glass production more energy efficient.

(ii) Increase in workability:- It improves the fluidity of molten glass, making it easier to shape into desired forms during manufacturing.

(iii) Cost-effective glass formation:- Soda ash is an inexpensive and widely available material, making it an economical choice for large-scale glass production.

Reaction:-



Soda ash reacts with silica to form sodium silicate and carbon dioxide gas. Sodium silicate acts as a

binding agent in glass structure, improving its workability and stability.

(iii)

Physically Tempered Safety Glass

Physically tempered safety glass, commonly known as tempered glass, is a type of safety glass that has been produced through controlled thermal or chemical treatments to enhance its strength compared to standard glass. This process introduces internal stresses that improve its mechanical properties and safety features, making it a preferred material in various industries.

Manufacturing Process

(i) Heating: The glass is uniformly heated to temperatures ranging from 620°C to 700°C , close to its softening point.

(ii) Rapid cooling (Quenching): The heated glass undergoes rapid cooling using high-pressure air jets. This sudden cooling creates a compressive stress layer on the outer surfaces, while the interior remains under tensile stress.

(iii) Resulting Structure: The balance of these stresses makes tempered glass 4-5 times stronger than standard annealed glass.

Properties

(i) Tempered glass is approximately four to five times stronger than standard glass, making it resistant to impacts and mechanical stress.

(ii) When broken, it fractures into small, granular pieces instead of sharp shards, reducing the risk of serious injury.

(iii) It can withstand higher temperatures and thermal shocks, making

(5)

it suitable for applications involving heat or sudden temperature changes.

Limitations

The glass must be cut and shaped before the tempering process, as it will shatter if altered afterward. While tempered glass is stronger than standard glass, it can still break under severe impacts or abrasion. Scratches by sharp objects in rare cases, nickel sulfide inclusions in the glass can cause spontaneous breakage.

(iv)

Surfactants classified. describe each class.

Surfactants, or surface-active agents, are compounds that lower the surface tension between two liquids or between a liquid and a solid. They are amphiphilic, meaning they contain both hydrophilic (water-loving) and hydrophobic (water-hating) parts. This dual nature allows surfactants to interact with both water and oil, making them essential in various applications such as detergents, emulsifiers, and wetting agents.

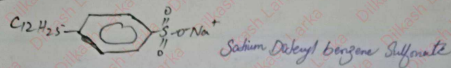
Classification:-

Surfactants are classified based on their solubility in water, which determines their ionic nature. The main classes of surfactants are:

(i) Anionic Surfactants: These surfactants dissociate

In water to form an amphiphilic anion and a cation usually an alkaline metal like sodium (Na^+) or potassium (K^+)

Example:-

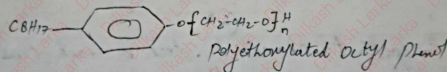


uses:- Commonly used in household and industrial cleaning products due to their strong cleaning and foaming properties.

(ii) Nonionic Surfactants:-

These surfactants do not ionize in water. Their hydrophilic part is typically a non-dissociable group like alcohol, phenol, ether, ester, or amide.

Example:-

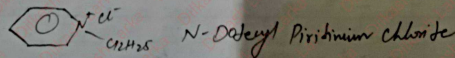


uses:- widely used in Cosmetics, Pharmaceuticals, and as emulsifiers in food products due to their milder and stability over a wide pH range.

(iii) Cationic Surfactants:-

These surfactants dissociate in water to form an amphiphilic cation and an anion, often a halide.

Example:-



uses:- Primarily used as antimicrobial agents, fabric softeners, and in hair conditioners due to their positive charge.

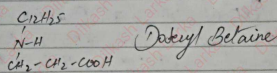
(7)

which allows them to bind to negatively charged surfaces.

(iv) Amphoterics (Zwitterionic) Surfactants:-

These surfactants contain both anionic and cationic groups within the same molecule, making them zwitterionic.

Example:-



uses:-

often used in personal care products like shampoos and body washes due to their compatibility with skin and low irritation potential.

(8)

Process involved in manufacturing of Maleic Anhydride from Naphthalene:-

Maleic Anhydride is an organic compound with the formula $\text{C}_4\text{H}_2(\text{O})_3$. It is the anhydride of maleic acid and is a colorless solid that plays a significant role in industrial chemistry. Its primary use is in the production of plasticizers, which are used to make plastic flexible. Maleic anhydride is also used in the production of polyester resins, alloy resins, dyes, insect repellents, and polyurethanes.

Preparation

(1) Melting and vaporization: Naphthalene is melted and pumped into a vaporizer. Primary preheated air is bubbled through the molten naphthalene to vaporize it. Additional air is mixed with the primary air-naphthalene vapor stream to achieve an air-naphthalene ratio of 18:22:12 by weight.

Preparation reactions:-

Repeat in Short Question 2023

(ii) Oxidation in the reactor: The vapor mixture is directed to a converter containing multiple tubes filled with vanadium pentoxide (V_2O_5) catalyst. The reactor temperature is maintained between 675-850°F. Naphthalene is oxidized to produce Phthalic anhydride, Carbon dioxide (CO_2), and Water (H_2O).

(iii) Cooling and Condensation: The exit gases pass through a vapor cooler, reducing the gas temperature to just above the dew point (about 260°F). Phthalic anhydride crystallizes on the walls of large air-cooled condensers.

(iv) Recovery and Refinement: The crude Phthalic anhydride is refined by distillation in a stainless steel vacuum still. The purified phthalic anhydride is melted and packed for storage and packaging.

(v) By product Removal:-

Maleic anhydride, a by product, is removed by processing the crystal in the last compartment of the condenser box system.

(9)

or by scrubbing the tail gases from condenser boxes.
Most commercial Phthalic anhydride contains 0.25-0.40 percent maleic anhydride.

flow chart:-

